



# Inductive Proofs

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Courtesy: slides are adopted from Discrete Mathematics and Its Applications by Kenneth H. Rosen

# Mathematical Induction

- A powerful, rigorous technique for proving that a predicate  $P(n)$  is true for *every* natural number  $n$ , no matter how large.
- Based on a predicate-logic inference rule:

- $$\frac{P(0) \quad \forall n \geq 0 (P(n) \rightarrow P(n+1))}{\therefore \forall n \geq 0 P(n)}$$



# Outline of an Inductive Proof

- *Base case* Want to prove  $\forall n P(n) \dots$
- *e* (or *basis step*): Prove  $P(0)$ .
- *Inductive step*: Prove  $\forall n P(n) \rightarrow P(n+1)$ .
  - e.g. use a direct proof:
    - Let  $n \in \mathbf{N}$ , assume  $P(n)$ . (*inductive hypothesis*)
    - Under this assumption, prove  $P(n+1)$ .
- Inductive inference rule then gives  $\forall n P(n)$ .

# Example

- $1 = (1 \times 2)/2$
- $1 + 2 = (2 \times 3)/2$
- $1 + 2 + 3 = (3 \times 4)/2$
- $1 + 2 + 3 + 4 = (4 \times 5)/2$
- $1 + 2 + 3 + 4 + 5 = 4 \times 5/2 + 5 = 5 \times 6/2$
- $P(n): 1 + 2 + \dots + n = n(n + 1)/2$
- Prove  $P(n + 1)$  under the assumption  $P(n)$  is true
- Indeed we prove  $\forall n \geq 1 (P(n) \rightarrow P(n + 1))$
- Or  $\forall n (P(n) \rightarrow P(n + 1))$  is tautology
- $1 + 2 + \dots + n + (n + 1) = n \times (n + 1)/2 + (n + 1) = (n + 1)(n + 2)/2$

# Validity of Induction

Prove: if  $\forall n \geq 0 (P(n) \rightarrow P(n+1))$ , then  $\forall k \geq 0 P(k)$

(a) Given any  $k \geq 0$ ,  $\forall n \geq 0 (P(n) \rightarrow P(n+1))$  implies

$$(P(0) \rightarrow P(1)) \wedge (P(1) \rightarrow P(2)) \wedge \dots \wedge (P(k-1) \rightarrow P(k))$$

(b) Using *hypothetical syllogism*  $k-1$  times we have  
 $P(0) \rightarrow P(k)$

(c)  $P(0)$  and *modus ponens* gives  $P(k)$ .

Thus  $\forall k \geq 0 P(k)$ .

# Another Induction Example

- Prove that  $\forall n \geq 0, n < 2^n$ . Let  $P(n) = (n < 2^n)$ 
  - Base case:  $P(0) = (0 < 2^0) = (0 < 1) = \text{T}$ .
  - Inductive step: For  $n \geq 0$  prove  $P(n) \rightarrow P(n+1)$ .
    - Assuming  $n < 2^n$ , prove  $n+1 < 2^{n+1}$ .
    - Note  $n + 1 < 2^n + 1$  (by inductive hypothesis)  
 $< 2^n + 2^n$  (because  $1 < 2 = 2 \cdot 2^0 \leq 2 \cdot 2^{n-1} = 2^n$ )  
 $= 2^{n+1}$
    - So  $n + 1 < 2^{n+1}$ , and we're done.

# The Well-Ordering Property

- The validity of the inductive inference rule can also be proved using the *well-ordering property*, which says:
  - Every non-empty set of non-negative integers has a minimum (smallest) element.
  - $\forall S \subseteq \mathbb{N} : \exists m \in S : \forall n \in S : m \leq n$

# Use Well-Ordering Property and Proof by Contradiction

- Suppose  $P(0)$  is true and that  $P(k) \rightarrow P(k+1)$  is true for all *positive*  $k$ .
- Assume  $P(n)$  is false for some positive  $n$ .
- Implies  $S = \{n \mid \neg P(n)\}$  is non-empty and has a minimum element  $m$  (i.e.,  $P(m) = \text{false}$ )
- But then,  $P(m-1) \rightarrow P((m-1)+1) = P(m)$  which is a contradiction!

# More Examples

Def:  $H_n = 1 + \frac{1}{2} + \dots + \frac{1}{n}$

Problem:  $H_{2^n} \geq 1 + \frac{n}{2}$

Solution:

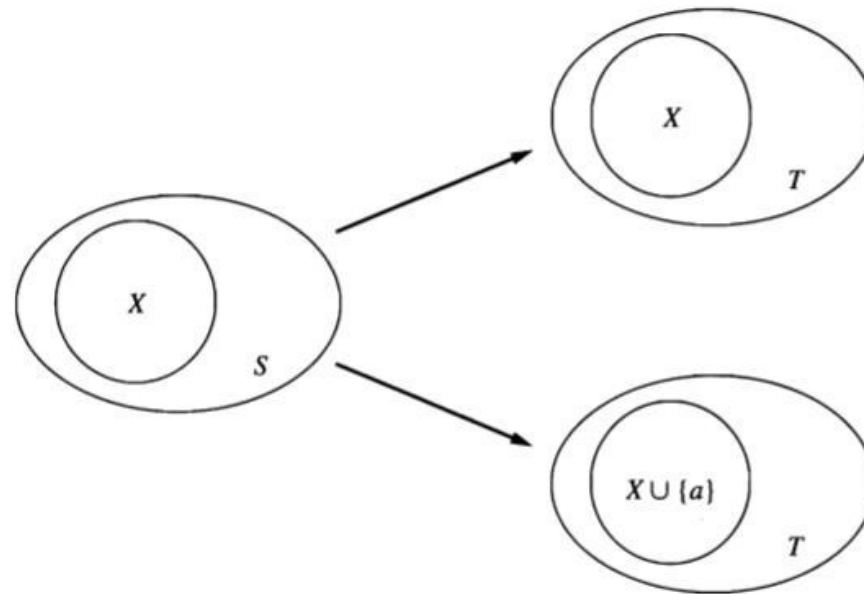
- Basis Step:  $1 + \frac{1}{2} \geq 1 + \frac{1}{2}$

- Inductive Step:  $H_{2^n} \geq 1 + \frac{n}{2} \quad H_{2^{n+1}} \geq 1 + \frac{n+1}{2}$

$$\begin{aligned} H_{2^{n+1}} &= H_{2^n} + \left( \frac{1}{2^n + 1} + \dots + \frac{1}{2^{n+1}} \right) \geq 1 + \frac{n}{2} + \frac{2^n}{2^{n+1}} \\ &= 1 + \frac{n+1}{2} \end{aligned}$$

# More Examples

- Problem: #subsets of  $T = 2^{|T|}$
- Solution:  $T = S \cup \{a\}$



# More Examples

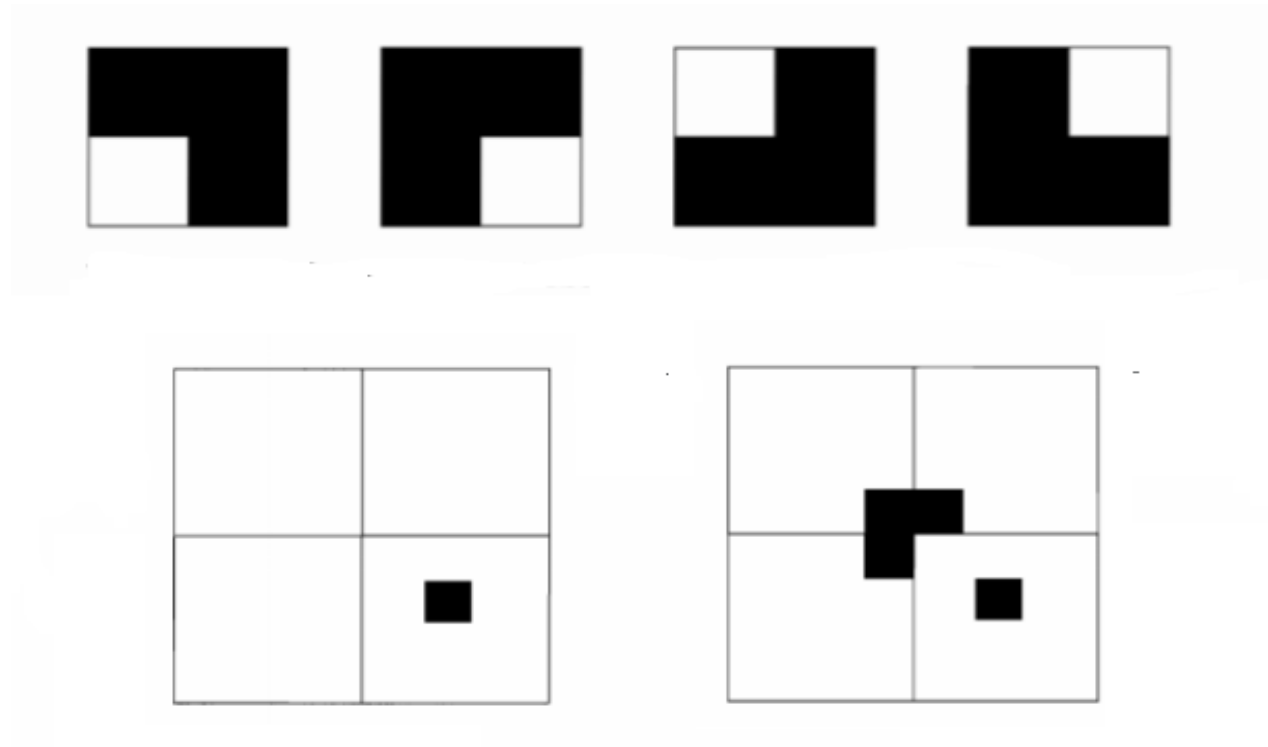
- Problem: De Morgan's laws  $\overline{\bigcap A_i} = \bigcup \overline{A_i}$

$$\begin{aligned}\overline{\bigcap_{j=1}^{k+1} A_j} &= \overline{\left(\bigcap_{j=1}^k A_j\right) \cap A_{k+1}} && \text{by the definition of intersection} \\ &= \overline{\left(\bigcap_{j=1}^k A_j\right) \cup \overline{A_{k+1}}} && \text{by De Morgan's law (where the two sets are } \bigcap_{j=1}^k A_j \text{ and } A_{k+1}) \\ &= \left(\bigcup_{j=1}^k \overline{A_j}\right) \cup \overline{A_{k+1}} && \text{by the inductive hypothesis} \\ &= \bigcup_{j=1}^{k+1} \overline{A_j} && \text{by the definition of union.}\end{aligned}$$

# More Examples

Problem: Tiling a  $2^n \times 2^n$  board with one arbitrary cell removed with right triominoes

Solution:



مثال: به ازای هر عدد صحیح  $n$  ثابت کنید:

$$1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1 \quad n \geq 0$$

پایه استوار:  $p(0): 2^{0+1} - 1 = 1$  جمله اول

گام استوار:  $p(k): 1 + 2 + 2^2 + \dots + 2^k = 2^{k+1} - 1$  فرض استوار

$$\begin{aligned} p(k+1): 1 + 2 + \dots + 2^k + 2^{k+1} &= 2^{k+1} - 1 + 2^{k+1} \\ &= 2^{k+2} - 1 \quad \square \end{aligned}$$

مثال: به ازای هر عدد طبیعی  $n$ ،  $n^3 - n$  بر 3 بخش پذیر است.

ثبات با استوار:

پایه استوار:  $p(1): 1 - 1 = 0 = 3k'$

گام استوار:  $k^3 - k = 3k'$

$$\begin{aligned} (k+1)^3 - (k+1) &= k^3 + 3k^2 + 3k + 1 - k \\ &= \underbrace{k^3 - k}_{3k'} + 3(k^2 + k) \\ &= 3k'' \quad \square \end{aligned}$$

مثال : به ازای هر عدد طبیعی  $n$  و  $x > 0$  داریم :

$$(1+x)^n \geq 1+nx$$

نشان بده از طریق استقراد :

پایه استقراد :  $P(1) : 1+x \geq 1+x \quad \checkmark$

گام استقراد :

$$(1+x)^n = (1+x)(1+x)^{n-1}$$

$$\geq (1+x)(1+(n-1)x)$$

$$= 1 + (n-1)x + x + (n-1)x^2$$

$$= 1 + nx + \cancel{x} + \cancel{x} + \frac{(n-1)x^2}{+} \geq 1+nx$$

مثال: به ازای هر  $n \geq 4$  داریم  $2^n < n!$   
اثبات به روش استقراء:

پایه استقراء:  $p(4): \frac{2^4}{16} < \frac{4!}{24}$

گام استقراء:  $p(n-1): 2^{n-1} < (n-1)!$

$2(2^{n-1}) < 2(n-1)! < n(n-1)! = n!$  طبق فرض

$p(n): 2^n < n!$  حکم  $\square$

مثال: به ازای هر  $n$  عدد طبیعی  $x$ ،  $x^n - 1$  بر  $x - 1$  بخش پذیر است

اثبات: با استقراء بر روی  $n$

$p(n-1): x^{n-1} - 1 = k(x-1)$   
فرض استقراء

$$x^n - 1 = x^n - x + x - 1$$

$$= x(\underbrace{x^{n-1} - 1}_{k(x-1)}) + (x-1) \quad \square$$

طبق فرض

$$\sum_{j=0}^n ar^j = a + ar + ar^2 + \dots + ar^n = \frac{ar^{n+1} - a}{r-1} \quad \text{مثال:}$$

$$r \neq 1$$

$$n \geq 0$$

نشان دهید که این فرمول برای هر عدد صحیح  $n$  (توسط روش استقرای ریاضی) درست است.

$$p(0) : \frac{ar^{0+1} - a}{r-1}$$

پایه استوار:  $n=0$

$$= \frac{ar - a}{r-1} = \frac{a(r-1)}{r-1} = a \quad \checkmark \text{ جمله اول}$$

گام استوار:

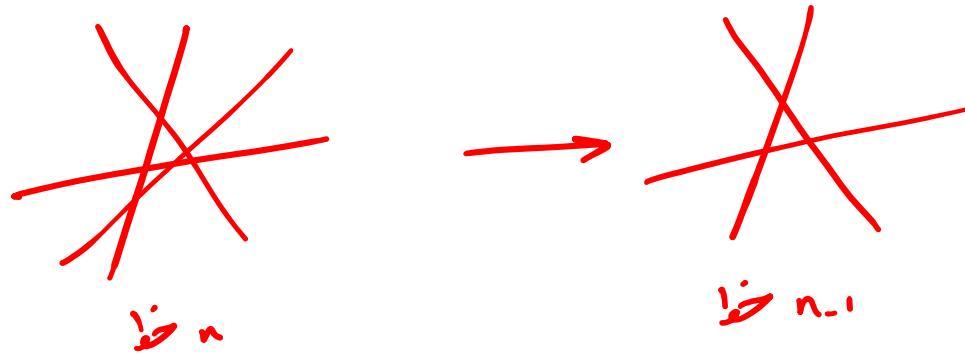
$$p(k) : a + ar + \dots + ar^k = \frac{ar^{k+1} - a}{r-1} \quad \text{فرض استوار}$$

$$p(k+1) : a + ar + \dots + ar^k + ar^{k+1} = \frac{ar^{k+1} - a}{r-1} + ar^{k+1}$$

$$= \frac{ar^{k+1} - a + ar^{k+2} - ar^{k+1}}{r-1}$$

$$= \frac{ar^{k+2} - a}{r-1} \quad \square$$

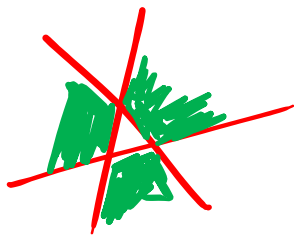
مثال:  $n$  خط، هر دو خط  $\frac{n(n+1)}{2} + 1$  ناحیه ای را می‌کند.



$$\underbrace{\frac{n(n-1)}{2} + 1}_{\text{فرض}} + n = \frac{n(n+1)}{2} + 1$$

مثال : نامه برای احمد شد، رستخوار خدا در صحنه را این زبان طوری با درشتی،

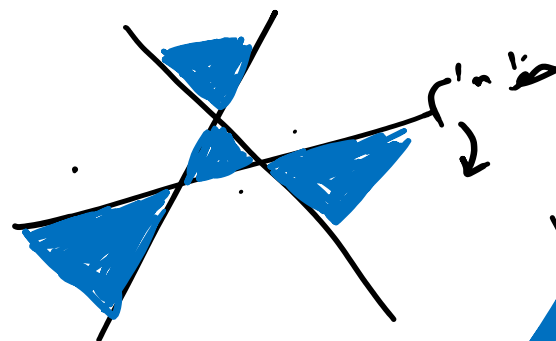
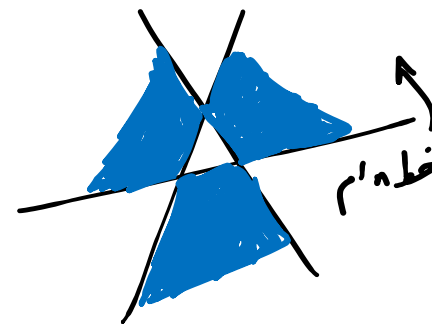
دُکّانِ اُمیرِ کرام، دُکّانِ حُجّیجِ دُرّنامیہ، محاورہ، حکمت، مابُشندہ .



in


$$h = 1$$
 $n = 2$ 

\* تغییر نیت از سر از ادب و خلعت ۸ ام کجاست !


$$\frac{n-1 \text{ خط (رضی)}}{n \text{ خط}}$$


# Strong Induction

- Characterized by another inference rule:

$$\frac{P(0) \quad \overbrace{P \text{ is true in all previous cases}} \quad \forall n \geq 0: (\forall 0 \leq k \leq n P(k)) \rightarrow P(n+1)}{\therefore \forall n \geq 0: P(n)}$$

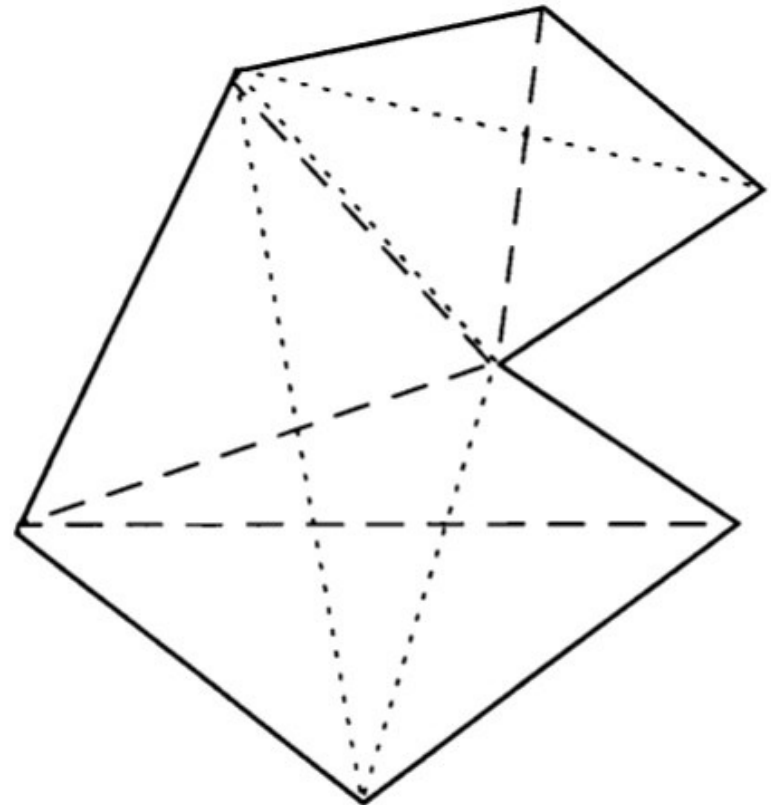
- Difference with previous version is that the inductive step uses the fact that ***P(k)*** is true **for *all* smaller  $k \leq n$ , not just for  $k=n$ .**

$$\underline{[P(1) \wedge P(2) \wedge P(3) \wedge \dots \wedge P(n)]} \rightarrow P(n+1)$$

# Example

Problem: A simple polygon with  $n$  vertices can be partitioned into triangles whose vertices are vertices of the simple polygon.

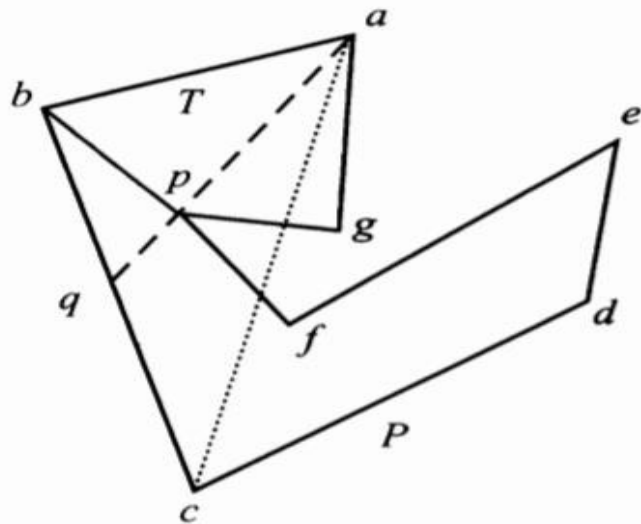
Two Triangulations  
Denoted by  
Dots and Dashes



# Example (Cont)

Solution:

- there is vertices  $p$  and  $q$  such that the segment  $pq$  is inside the polygon.
- Decompose the polygon into two polygon with fewer vertices by drawing  $pq$
- Triangulate each subpolygon



$T$  is the triangle  $abc$

$p$  is the vertex of  $P$  inside  $T$  such that the  $\angle bap$  is smallest

$bp$  must be an interior diagonal of  $P$

# Faulty Proof

Problem:  $P(n)$ : Any set of  $n$  non-parallel lines meet in a common point

Solution:

- Basis Step:  $P(2)$  is true
- Inductive Step:  $P(k) \rightarrow P(k+1)$

$\{l_1, l_2, \dots, l_{k-1}, l_k\}$  meet at intersection of  $l_1, l_2$

$\{l_1, l_2, \dots, l_{k-1}, l_{k+1}\}$  meet at intersection of  $l_1, l_2$

Then  $\{l_1, l_2, \dots, l_{k-1}, l_k, l_{k+1}\}$  meet at intersection of  $l_1, l_2$

Above argument is true when  $k$  is at least 3. Then  $P(3)$  is basis step which of course is false.

# Functions

## Fibonacci Numbers

- $f(n+1) = f(n) + f(n-1), f(1) = f(2) = 1$

Problem:  $f(n) > \alpha^{n-2}$  where  $\alpha = (\sqrt{5} + 1)/2$

Solution:

$$\begin{aligned} f(n+1) &= f(n) + f(n-1) > \alpha^{n-2} + \alpha^{n-3} \\ &= \alpha^{n-3}(\alpha + 1) = \alpha^{n-3}\alpha^2 = \alpha^{n-1} \end{aligned}$$

# Example

- Show that every  $n > 1$  can be written as a product  $p_1 p_2 \dots p_s$  of some series of  $s$  prime numbers. Let  $P(n)$  = " $n$  has that property"
- Base case:  $n=2$ , let  $s=1, p_1=2$ .  
 $n = 2 = 2$  بار استقر
- Inductive step: Let  $n \geq 2$ . Assume  $\forall 2 \leq k \leq n: P(k)$ .  
فرض استقر  
 Consider  $n+1$ . If prime, let  $s=1, p_1=n+1$ . ✓  
 Else  $n+1=ab$ , where  $1 < a \leq n$  and  $1 < b \leq n$ .  
 Then  $a = p_1 p_2 \dots p_t$  and  $b = q_1 q_2 \dots q_u$ . Then  $n+1 = p_1 p_2 \dots p_t q_1 q_2 \dots q_u$ , a product of  $s=t+u$  primes. ✗

# Example

Suppose  $a_0, a_1, a_2, \dots$  is defined as follows:

$$\underline{a_0=1}, \underline{a_1=2}, \underline{a_2=3},$$

$$\underline{a_k = a_{k-1} + a_{k-2} + a_{k-3}} \text{ for all integers } k \geq 3.$$

Then  $\underline{a_n \leq 2^n}$  for all integers  $n \geq 0$ .  $P(n)$

➤ **Proof** (*by strong induction*):

1) **Basis step:**

The statement is true for  $n=0$ :  $a_0=1 \leq 1=2^0$  ✓  $P(0)$

for  $n=1$ :  $a_1=2 \leq 2=2^1$  ✓  $P(1)$

for  $n=2$ :  $a_2=3 \leq 4=2^2$  ✓  $P(2)$

# Example (cont'd)

2) Inductive hypothesis: For any  $k > 2$ ,

Assume  $P(i)$  is true for all  $i$  with  $0 \leq i < k$ :

$$a_i \leq 2^i \text{ for all } 0 \leq i < k.$$

3) Inductive step:

Show that  $P(k)$  is true:  $a_k \leq 2^k$

$$a_k = a_{k-1} + a_{k-2} + a_{k-3}$$

$$\leq 2^{k-1} + 2^{k-2} + 2^{k-3}$$

(using inductive hypothesis)

$$\leq 2^0 + 2^1 + \dots + 2^{k-3} + 2^{k-2} + 2^{k-1}$$

$$= 2^k - 1$$

(as a sum of geometric sequence)

$$\leq 2^k$$

□

Thus,  $P(n)$  is true by strong induction.

# Sets

- Basis Step:  $3 \in S$
- Recursive Step:  $x \in S, y \in S \rightarrow x + y \in S$

مجموعه  
تکراری

→ Problem: Prove  $S = \{3k | k \in \mathbb{N}\}$

- $A = \{3k | k \in \mathbb{N}\}$
- $3 \in S, 3k \in S \rightarrow 3(k+1) \in S$

• Then  $A \subseteq S$

• Prove  $S \subseteq A$

تکرار

همه اعداد 3 مضرب هستند

همه اعداد 3 مضرب هستند

# String

Defining  $\Sigma^*$

- Basis Step:  $\lambda \in \Sigma^*$
- Recursive Step:  $x \in \Sigma^*, a \in \Sigma \rightarrow xa \in \Sigma^*$

Defining the length

- $L(\lambda) = 0$
- $L(xa) = L(x) + 1$

Problem:  $L(xy) = L(x) + L(y)$

- Basis:  $L(x\lambda) = L(x) + L(\lambda) = L(x)$  ✓
- Inductive Step:  $y = wa, w \in \Sigma^*, a \in \Sigma$   

$$L(xy) = L(xwa) = L(xw) + 1 = L(x) + L(w) + 1$$

$$= L(x) + L(y)$$

تاریخچه این مسئله  
از ترمینال به زبان

حروف الفبا

استند. طول

در طول

فرض استند

# Propositional expression

- Basis Step: T, F, and  $s$  (a propositional variable)
- Recursive Step:

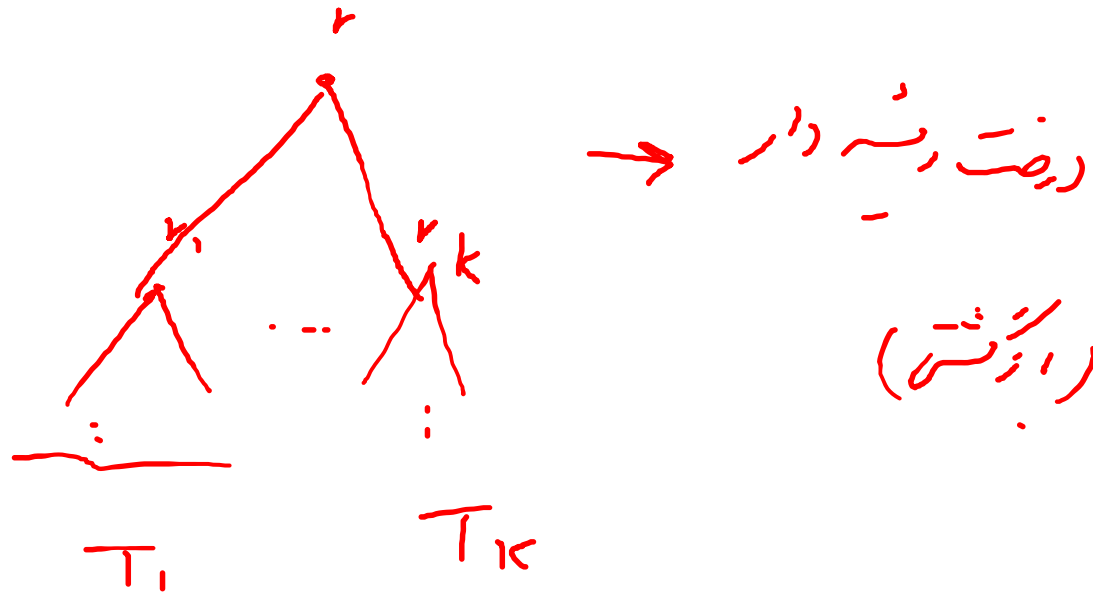
$$\underline{E}, \underline{F} \rightarrow (\sim \underline{E}), (\underline{E} \rightarrow \underline{F}), (\underline{E} \wedge \underline{F}), (\underline{E} \vee \underline{F}), (\underline{E} \leftrightarrow \underline{F})$$

← معنای استنتاج

Problem: Show any expression contains an equal number of left and right parentheses

# Rooted Trees

- Basis Step: A single node is a rooted tree
- Recursive Step: If  $T_1, T_2, \dots, T_k$  are rooted trees with roots of  $r_1, r_2, \dots, r_k$ , then a tree with root  $r$  and edges from  $r$  to  $\overline{r_1}, \overline{r_2}, \dots, \overline{r_k}$  is a rooted tree



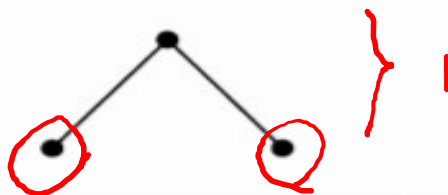
# Fully Binary Tree

- Basis Step: A single node is a fully binary tree

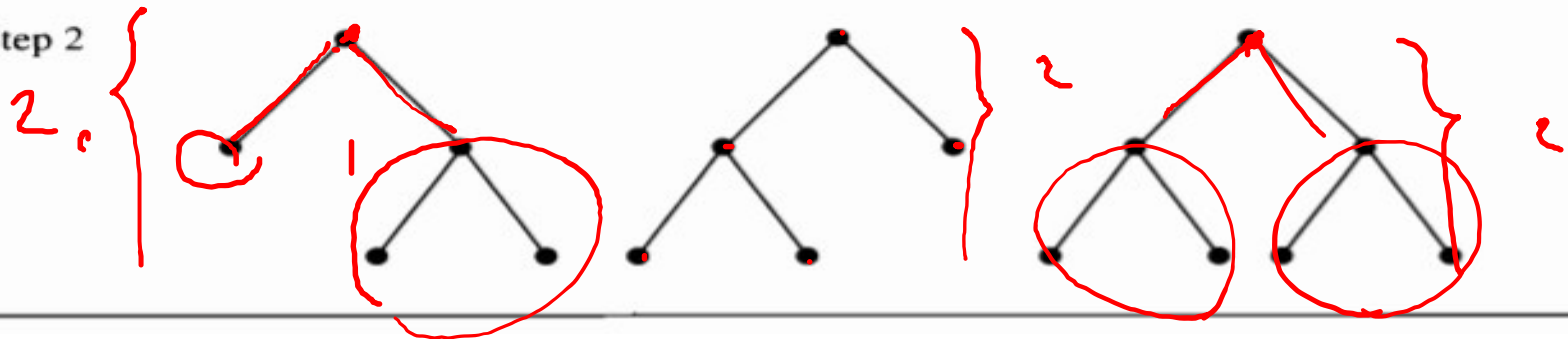
Basis step



Step 1



Step 2



# Full Binary Tree

Height ( $h(T)$ )

- Basis Step: If  $T$  is a single node,  $h(T)=0$
- Recursive Step: If  $T$  constructed from  $T_1, T_2$ , then  
 $h(T)=\max(h(T_1), h(T_2))+1$

Size or the number of nodes ( $n(T)$ )

Problem:  $n(T) \leq 2^{h(T)+1} - 1$

وضوح الشرح  
 $T_1, T_2$

$$\begin{aligned}
 n(T) &= 1 + n(T_1) + n(T_2) \\
 &\leq 1 + (2^{h(T_1)+1} - 1) + (2^{h(T_2)+1} - 1) \\
 &\leq 1 + (2^{h(T)} - 1) + (2^{h(T)} - 1) = (2^{h(T)+1} - 1) \quad \square
 \end{aligned}$$

# Multi-variables

For Fibonacci Num:

$$F(n + m) = F(n + 1)F(m) + F(n)F(m - 1)$$

Fibonacci Num:  $F(n + 1) = F(n) + F(n - 1),$

$$F(1) = F(2) = 1, F(0) = 0$$

Fix  $n$  and induction on  $m$

- Basis Step:  $m = 1$

$$F(n + 1) = F(n + 1)F(1) + F(n)F(0) = F(n + 1) \quad \checkmark$$

- Inductive Step

$$\begin{aligned} F(n + m) &= F((n + 1) + (m - 1)) \\ &= F(n + 2)F(m - 1) + F(n + 1)F(m - 2) \\ &= (F(n + 1) + F(n))F(m - 1) + F(n + 1)F(m - 2) \\ &= F(n + 1)(F(m - 1) + F(m - 2)) + F(n)F(m - 1) \\ &= F(n + 1)F(m) + F(n)F(m - 1) \quad \square \end{aligned}$$

ف.ب.  $\square$

# Muti-Variables

For Fibonacci Num:

$$F(n + m) = F(n + 1)F(m) + F(n)F(m - 1)$$

- induction on  $n + m$

- Basis Step:

$$n = \dots, m = 1$$

$$n = 1, m = \dots$$

For  $n + m < 2$  check

$$F(n + m) = F(n + 1)F(m) + F(n)F(m - 1)$$

- Inductive Step:

$$\begin{aligned}
 F(n + m) &= F(n + m - 1) + F(n + m - 2) \\
 &= F(n + (m - 1)) + F(n + (m - 2)) \\
 &= F(n + 1)F(m - 1) + F(n)F(m - 2) \\
 &\quad + F(n + 1)F(m - 2) + F(n)F(m - 3) \\
 &= F(n + 1)(F(m - 1) + F(m - 2)) \\
 &\quad + F(n)(F(m - 2) + F(m - 3)) \\
 &= F(n + 1)F(m) + F(n)F(m - 1)
 \end{aligned}$$

فیب. سؤفت

فیب. سؤفت

